open data | data on request | closed data | preprint | reanalysis on open data | our reproduction

(a) Maximum vocabulary size used in the 32 language-BCI studies

vocabulary size

71,400 word forms (Brysbaert et al., 2016)
42,000 lemmas (Brysbaert et al., 2016)
11,100 word families (Brysbaert et al., 2016)

Angrick et al., 2024 (CCO1)	Wandelt et al., 2024 (P1/P2)	Brosler et al., 2026 (B1r/B3/B6)	Moses et al., 2021 (B1)	Silva et al., 2024b (B1)	Metzger et al., 2023 (B3)	Littlejohn et al., 2025 (B3)	Metzger et al., 2022 (B1)	Feghhi et al., 2026 (T12/T15)	Willett et al., 2023 (T12)	Card et al., 2024 (T15)	Feng et al., 2024 (T12)	Kunz et al., 2025 (T12/T15/T16)	Li et al., 2025 (T12)	Feghhi et al., 2025 (T12)	Levin et al., 2026 (T12/T15/T16)	Y. Zhang et al., 2026 (T12/T15)	Nason-Tomaszewski et al., 2026 (T16)	Boccato et al., 2026 (T12/T15/T16/T17)	Jude et al., 2026a (T17/T18)	Jude et al., 2026b (T17)	Yoon et al., 2026 (T12)	Card et al., 2026 (T15)	Fogg et al., 2026 (T12/T15/T16/T17/T21/T22)	Z. Zhang et al., 2026 (T12/T15)	Pouratian et al., 2026 (V1/V2/V3)	Fan et al., 2023 (T5)	Karpowicz et al., 2024 (T5)	Willett et al., 2021 (T5)	Wairagkar et al., 2025 (T15)	Levin et al., 2026 (T5/T12/T18)	Wairagkar et al., 2026 (T15)
6	8	5–10	50	178	1,024	1,024	1,152	100k	125k	125k	125k	125k	125k	125k	125k	125k	125k	125k	125k	125k	125k	125k	125k	125k	125k	125k	130k	130k	open (brain-to-text without language model)	open (brain-to-voice)	open (brain-to-text without language model)	open (brain-to-voice)

Filtered to 8 open-data studies exceeding native-speaker vocabulary (>71,400 word forms)

(b) Open-sourced dataset size

released sentences

Willett et al., 2021 (T5)	Willett et al., 2023 (T12)	Fan et al., 2023 (T5)	Card et al., 2024 (T15)	Wairagkar et al., 2025 (T15)	Kunz et al., 2025 (T12)	Kunz et al., 2025 (T15)	Kunz et al., 2025 (T16)	Jude et al., 2026a (T17)	Jude et al., 2026a (T18)	Jude et al., 2026b (T17)
10 d/sess, 95 blks, span 253 d	24 d/sess, 275 blks, span 120 d	21 d/sess, 105 blks, span 287 d	45 d/sess, 265 blks, span 612 d	37 d/sess, blks n/a, span 171 d	1 d/sess, 12 blks, span 1 d	3 d/sess, 18 blks, span 309 d	2 d/sess, 14 blks, span 234 d	8 d/sess, 46 blks, span 89 d	9 d/sess, 98 blks, span 119 d	13 d/sess, 96 blks, span 141 d
1,000	9,680	1,315	9,498	8,489	500	720	450	668	1,033	3,674

(c) Word error rate (online)
no series printed: Wairagkar et al., 2025 (listener median)

† recomputed from released data (decode outputs / trial timing / Source Data files)
‡ digitized from the paper's figure markers
error bar is calculated across days

word error rate (%)

12.8% 14-dataset average (Radford et al., 2022)
5.8% Switchboard (Xiong et al., 2017)
2.5% LibriSpeech (Radford et al., 2022)

Willett et al., 2021 (T5)	Willett et al., 2023 (T12)	Fan et al., 2023 (T5)	Card et al., 2024 (T15)	Wairagkar et al., 2025 (T15)	Kunz et al., 2025 (T12)	Kunz et al., 2025 (T15)	Kunz et al., 2025 (T16)	Jude et al., 2026a (T17)	Jude et al., 2026a (T18)	Jude et al., 2026b (T17)
vocab open mean±SD† 5 of 10 d: 25.1±8.6	vocab 125k mean±SD† 5 of 24 d: 23.9±1.8	vocab 130k mean±SD† 10 of 21 d: 4.0±1.2	vocab 125k mean±SD† 14 of 45 d: 4.9±3.3	vocab open listener median 8 of 37 d: 43.8	vocab 50-w mean±SD† 1 of 1 d: 23.7±0.0	vocab 125k mean±SD† 2 of 3 d: 51.3±33.1	vocab 125k mean±SD† 1 of 2 d: 46.6±0.0	vocab 125k mean±SD† 6 of 8 d: 12.5±2.5	vocab 125k mean±SD† 5 of 6 d: 3.4±0.8	vocab 125k mean±SD† 4 of 13 d: 66.5±18.1

(d) Communication speed (online)
n/r: speed not reported

† recomputed from released data (decode outputs / trial timing / Source Data files)
‡ digitized from the paper's figure markers
error bar is calculated across days

words per minute

160 wpm natural conversation (Card et al., 2024)
51.6 wpm keyboard typing (Dhakal et al., 2018)
36.2 wpm smartphone typing (Palin et al., 2019)
8 wpm point-and-click BCI (Willett et al., 2021)
5–10 wpm eye typing (Hamid & Kristensson, 2024)

Willett et al., 2021 (T5)	Willett et al., 2023 (T12)	Fan et al., 2023 (T5)	Card et al., 2024 (T15)	Wairagkar et al., 2025 (T15)	Kunz et al., 2025 (T12)	Kunz et al., 2025 (T15)	Kunz et al., 2025 (T16)	Jude et al., 2026a (T17)	Jude et al., 2026a (T18)	Jude et al., 2026b (T17)
vocab open mean±SD† 5 of 10 d: 15.4±2.2	vocab 125k mean±SD† 5 of 24 d: 65.3±0.9	vocab 130k mean±SD† 10 of 21 d (10 released): 14.6±2.2	vocab 125k mean±SD† 14 of 45 d: 32.5±2.7	n/r	n/r	n/r	n/r	vocab 125k mean±SD† 6 of 8 d: 6.8±1.8	vocab 125k mean±SD† 7 of 9 d: 14.0±4.5	vocab 125k mean±SD† 8 of 13 d: 34.1±7.7

Our reproduction on a 64 GB RAM / RTX 5060 Ti (16 GB) home PC (8 open-data studies)

(e) Reproduction: acoustic / synthesis stage (no LM), one pair per panel-(c) bar

our reproduction (16 GB home PC)
paper / benchmark anchor

Greedy PER / CER / 1−r (%)

Willett et al., 2021 (T5, raw CER)	Willett et al., 2023 (T12, greedy PER)	Fan et al., 2023 (T5, CORP vo CER)	Card et al., 2024 (T15, greedy PER)	Wairagkar et al., 2025 (T15, 1−mel r ²%)	Kunz et al., 2025 (T12, greedy PER)	Kunz et al., 2025 (T15, greedy PER)	Kunz et al., 2025 (T16, greedy PER)	Jude et al., 2026a (T17, raw CER)	Jude et al., 2026a (T18, raw CER)	Jude et al., 2026b (T17, greedy PER)
4.7 / 2.8	23.4 / 19.7	seed: 2.65; 6.2 / 6.2	10.2 / 9.5	r: 0.69 vs 0.83; 31.4 / 17.0	50-w sess.; 61.4 / 24.0	3-sess. subset; 59.7 / 54.0	2-sess. subset; 61.5 / 52.0	best; SrcData; 16.5 / 18.6	paper <12%; 5.3 / 12.0	40.0 / 34.0

(f) Reproduction: full-pipeline WER, one pair per panel-(c) bar

our reproduction (16 GB home PC)
paper / benchmark anchor

Word error rate (%)

Willett et al., 2021 (T5)	Willett et al., 2023 (T12)	Fan et al., 2023 (T5)	Card et al., 2024 (T15)	Wairagkar et al., 2025 (T15)	Kunz et al., 2025 (T12)	Kunz et al., 2025 (T15)	Kunz et al., 2025 (T16)	Jude et al., 2026a (T17)	Jude et al., 2026a (T18)	Jude et al., 2026b (T17)
bigram 63-tr. pool; 10.0 / 3.6	day med. 5-gram; 23.6 / 23.8	recal. arm 10 sess.; 4.1 / 6.2	41-s pool biGRU-48k; 6.2 / 6.7	WER n/r (voice; see e)	50-w sess. gap attr.; 94.3 / 24.0	125k gap attr.; 78.1 / 54.0	125k gap attr.; 82.4 / 52.0	best sess.; 11.5 / 15.0	best; paper <7%; 5.7 / 7.0	conv. day-68; 44.4 / 50.0

(g) Host RAM walls: official stacks vs. our 64 GB PC

official / naive stack
our consumer-PC stack
64 GB our host RAM

Peak host RAM (GB)

125k 5-gram WFST LM (host RAM; 4 studies)	125k 3-gram WFST LM (host RAM; 3 studies)	Typing 5-gram WFST (host RAM; Jude 2026a)	Handwriting fold + FST (host RAM; Willett 2021)
300 → 19.6 GB (mmap + .idx; 15.3×)	40–60 → 1.46 GB (KenLM trie; 41×)	64+ → 22.5 GB (2.8×) (drop unpruned FST)	56 → 26.9 GB (2.1×) (mmap + lazy FST)

(h) GPU VRAM: official rigs vs. our single 16 GB GPU

official provisioned GPU VRAM
our peak VRAM (1× RTX 5060 Ti)
16 GB our GPU VRAM

Peak GPU VRAM (GB)

Speech RNN + OPT-6.7B (A100 80G; Willett 2023)	Speech RNN + OPT-6.7B (1–2× 4090; 3 studies)	Voice synthesis (3× 3090; Wairagkar 2025)	Handwriting GRU + GPT-2 (1080 Ti / A100; 2 studies)	Typing GRU + 5-gram (RTX 4090; Jude 2026a)
80 GB A100 → 13.0 GB (OPT fp16)	24–48 GB 4090 → 13.0 GB (accum + OPT)	3×24 GB 3090 + A6000 → 0–1.5 GB (CPU / .h5)	40–44 GB rig → 3.5–4.0 GB (GRU + GPT-2)	24 GB 4090 → 11.5 GB (bf16 port)

(i) Session-level PER vs. post-LM WER

No-LM baseline: $1-(1-\mathrm{PER})^{3.4}$
WER = PER

word error rate after LM stage (%)
greedy phoneme error rate per session (%)

PER = 25%
(≤25%: 45/57 = 79% WER < PER;
>25%: 4/20 = 20% WER < PER)
This means that we must get PER < 25%.

per-session (open) pooled (star)
Willett et al., 2023 (T12)
Card et al., 2024 (T15)
Jude et al., 2026b (T17)